

LAB ASSIGNMENT 2



COMSATS University Islamabad

Programming Fundamentals

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QUESTIONS :

Question – 1:

Write a program that prompts the user to enter a Social Security number in the format DDD-DD-DDDD, where D is a digit. Your program should check whether the input is valid. Here are sample runs:

Enter a SSN: 232-23-5435

232-23-5435 is a valid social security number

Enter a SSN: 23-23-5435

23-23-5435 is an invalid social security number

Question – 2:

Write a program that prompts the user to enter two characters and displays the major and status represented in the characters. The first character indicates the major and the second is number character 1, 2, 3, 4, which indicates whether a student is a freshman, sophomore, junior, or senior. Suppose the following characters are used to denote the majors:

M: Mathematics
Technology

C: Computer Science

I: Information

Question – 3:

Assume a vehicle plate number consists of three uppercase letters followed by four digits. Write a program to generate a plate number.

Question – 4:

Write a program that prompts the user to enter a decimal integer and displays its corresponding binary value. Don't use Java's Integer.toString(int) in this program.

Question – 5:

The controversial issue of global warming has been widely publicized by the film “An Inconvenient Truth,” featuring former Vice President Al Gore. Mr. Gore and a U.N. network of scientists, the Intergovernmental Panel on Climate Change, shared the 2007 Nobel Peace Prize in recognition of “their efforts to build up and disseminate greater knowledge about man-made climate change.” Research both sides of the global warming issue online (you might want to search for phrases like “global warming skeptics”). Create a five-question multiple choice quiz on global warming, each question having four possible answers (numbered 1–4). Be objective and try to fairly represent both sides of the issue. Next, write an application that administers the quiz, calculates the number of correct answers (zero through five) and returns a message to the user. If the user correctly

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answers five questions, print "Excellent"; if four, print "Very good"; if three or fewer, print "Time to brush up on your knowledge of global warming," and include a list of some of the websites where you found your facts

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SOURCE CODE

assignment2 - Notepad

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```
import java.util.Scanner;
public class assignment2
{
    static Scanner input = new Scanner(System.in);
    public static void main(String [] args)
    {
        int option;
        do
        {
            System.out.println("-----> MAIN MENU (ASSIGNMENT - 2) <-----");

            System.out.println("1 - QUESTION 1");
            System.out.println("2 - QUESTION 2");
            System.out.println("3 - QUESTION 3");
            System.out.println("4 - QUESTION 4");
            System.out.println("5 - QUESTION 5");

            System.out.println("Press Option (1 - 5) to execute the program and press 0 to Exit");
            System.out.println("Enter Your Option: ");
            option = input.nextInt();

            switch(option)
            {
                case 1:
                    Q1();
                    break;
                case 2:
                    Q2();
                    break;
                case 3:
                    Q3();
                    break;
                case 4:
                    Q4();
                    break;
                case 5:
                    Q5();
```

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```
        Q5();
        break;
    case 0:
        System.out.println("Exiting The Program");
        return;
    default:
        System.out.println("Enter between 1 to 5");
    }
}
while(option != 0);
}
public static void Q1()
{
    boolean repeat = true;
    String ssn;
    do
    {
        System.out.println("-----> EXECUTING QUESTION - 1 <-----");

        System.out.println("Enter A SSN : ");
        ssn = input.next();
        if (validSSN(ssn))
        {
            System.out.println(ssn + " Is A Valid SSN");
        }
        else
        {
            System.out.println(ssn + " Is Not A Valid SSN");
        }
        System.out.println("Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu");
        System.out.println("Enter your choice : ");
        char choice = input.next().charAt(0);

        if (choice == 'y' || choice == 'Y')
        {
            repeat = true;
        }
    }
}
```

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```
        if (choice == 'y' || choice == 'Y')
        {
            repeat = true;
        }
        else if ((choice == 'b' || choice == 'B') || choice == '0')
        {
            repeat = false;
        }
        else
        {
            System.out.println("Invalid entry Returning to main menu");
            repeat = false;
        }
    }
    while (repeat);
}
public static boolean validSSN(String ssn)
{
    if (ssn.length() != 11)
    {
        return false;
    }
    else
    {
        for (int i = 0; i < ssn.length(); i++)
        {
            if ( i == 3 || i == 6 )
            {
                if ( ssn.charAt(i) != '-')
                {
                    return false;
                }
            }
            else
            {
                if (!Character.isDigit(ssn.charAt(i)))
                {

```

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```
        else
        {
            if (!Character.isDigit(ssn.charAt(i)))
            {
                return false;
            }
        }
    }
    return true;
}

}

public static void Q2()
{
    boolean repeat = true;
    do
    {
        String major = "";
        String status = "";

        System.out.println("-----> EXECUTING QUESTION - 2 <-----");

        System.out.println("Enter two Characters : ");
        String ms = input.next();
        if ( ms.length() != 2 )
        {
            System.out.println("Invalid Entry");
        }
        else
        {
            char A = ms.charAt(0);
            char B = ms.charAt(1);

            switch (A)
            {
                case 'M':
                case 'm':
                    major = "Mathmatics";
            }
        }
    }
    while (repeat);
}
```

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```
        major = "Mathmatics";
        break;
    case 'C':
    case 'c':
        major = "Computer Science";
        break;
    case 'I':
    case 'i':
        major = "Information Technology";
        break;
    default:
        major = "Invalid";
        break;
}

switch (B)
{
    case '1':
        status = "Freshman";
        break;
    case '2':
        status = "Sophomore";
        break;
    case '3':
        status = "Junior";
        break;
    case '4':
        status = "Senior";
        break;
    default:
        status = "Invalid";
        break;
}

}
if (major == "Invalid" || status == "Invalid")
{
```

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```
boolean repeat = true;
do
{
    String plate = "";

    System.out.println("-----> EXECUTING QUESTION - 3 <-----");

    for (int i = 0; i < 3; i++)
    {
        char letter = (char)(65 + (int)(Math.random() * 26));
        plate = plate + letter;
    }
    for (int i = 0; i < 4; i++)
    {
        int digits = (int)(Math.random() * 10);
        plate = plate + digits;
    }
    System.out.println("Plate Number : " + plate);

    System.out.println("Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu");
    System.out.println("Enter your choice : ");
    char choice = input.next().charAt(0);

    if (choice == 'y' || choice == 'Y')
    {
        repeat = true;
    }
    else if ((choice == 'b' || choice == 'B') || choice == '0')
    {
        repeat = false;
    }
    else
    {
        System.out.println("Invalid entry Returning to main menu");
        repeat = false;
    }
}
```

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```
repeat = false;
    }
    else
    {
        System.out.println("Invalid entry Returning to main menu");
        repeat = false;
    }
}
while (repeat);
}
public static void Q4()
{
    boolean repeat = true;
    do
    {
        String binary = "";
        String newbinary = "";

        System.out.println("-----> EXECUTING QUESTION - 4 <-----");

        System.out.println("Enter A Decimal Integer : ");
        int decint = input.nextInt();
        if ( decint == 0)
        {
            System.out.println("Binary Form = 0");
            return;
        }
        int num = decint;
        while (num > 0)
        {
            int remainder = num % 2;
            binary = binary + remainder;
            num /= 2;
        }
        for (int i = binary.length() - 1; i >= 0; i--)
        {
            char bin = binary.charAt(i);
```

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assignment2 - Notepad

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```
    }
    for (int i = binary.length() - 1; i >= 0; i--)
    {
        char bin = binary.charAt(i);
        newbinary = newbinary + bin;
    }
    System.out.println("Binary Form = " + newbinary);

    System.out.println("Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu");
    System.out.println("Enter your choice : ");
    char choice = input.next().charAt(0);

    if (choice == 'y' || choice == 'Y')
    {
        repeat = true;
    }
    else if ((choice == 'b' || choice == 'B') || choice == '0')
    {
        repeat = false;
    }
    else
    {
        System.out.println("Invalid entry Returning to main menu");
        repeat = false;
    }
}
while (repeat);
}
public static void Q5()
{
    boolean repeat = true;
    int score = 0;
    do
    {
        System.out.println("-----> EXECUTING QUESTION - 5 <-----");

        System.out.println("----- QUIZ -----");
```

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```
System.out.println("----- QUIZ -----");

System.out.println("1. The Earth's average temperature has increased over the past century.");
System.out.println("a. True\t\tb. False\t\tc. Not sure\t\td. None");
char q1 = input.next().charAt(0);
if (q1 == 'a' || q1 == 'A')
{
    score++;
}
System.out.println("2. The main cause of global warming is deforestation.");
System.out.println("a. True\t\tb. False\t\tc. Not sure\t\td. None");
char q2 = input.next().charAt(0);
if (q2 == 'b' || q2 == 'B')
{
    score++;
}
System.out.println("3. Renewable energy sources can help reduce global warming.");
System.out.println("a. True\t\tb. False\t\tc. Not sure\t\td. None");
char q3 = input.next().charAt(0);
if (q3 == 'a' || q3 == 'A')
{
    score++;
}
System.out.println("4. Global warming effects are only temporary.");
System.out.println("a. True\t\tb. False\t\tc. Not sure\t\td. None");
char q4 = input.next().charAt(0);
if (q4 == 'b' || q4 == 'B')
{
    score++;
}
System.out.println("5. The polar ice caps are melting due to global warming.");
System.out.println("a. True\t\tb. False\t\tc. Not sure\t\td. None");
char q5 = input.next().charAt(0);
if (q5 == 'a' || q5 == 'A')
{
    score++;
}
```

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```
}
System.out.println("\nYour score: " + score + " out of 5");
if (score == 5)
{
    System.out.println("Excellent");
}
else if (score == 4)
{
    System.out.println("Very good");
}
else
{
    System.out.println("Time to brush up on your knowledge of global warming.");
    System.out.println("Some helpful websites to learn more:");
    System.out.println("- https://climate.nasa.gov/");
    System.out.println("- https://www.epa.gov/climate-change");
    System.out.println("- https://www.ipcc.ch/");
    System.out.println("- https://climate.nationalgeographic.com/");
}

System.out.println("Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu");
System.out.println("Enter your choice : ");
char choice = input.next().charAt(0);
if (choice == 'y' || choice == 'Y')
{
    repeat = true;
}
else if ((choice == 'b' || choice == 'B') || choice == '0')
{
    repeat = false;
}
else
{
    System.out.println("Invalid entry Returning to main menu");
    repeat = false;
}
}
```

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```
        }  
    }  
    while (repeat);  
}  
<
```

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OUTPUT

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6396]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\assignment 2 pf>javac assignment2.java

C:\Users\shehyar\Desktop\assignment 2 pf>java assignment2.java
-----> MAIN MENU (ASSIGNMENT - 2) <-----
1 - QUESTION 1
2 - QUESTION 2
3 - QUESTION 3
4 - QUESTION 4
5 - QUESTION 5
Press Option (1 - 5) to execute the program and press 0 to Exit
Enter Your Option:
1
-----> EXECUTING QUESTION - 1 <-----
Enter A SSN :
234-57-9582
234-57-9582 Is A Valid SSN
Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu
Enter your choice :
0
```

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OUTPUT

```
-----> MAIN MENU (ASSIGNMENT - 2) <-----
1 - QUESTION 1
2 - QUESTION 2
3 - QUESTION 3
4 - QUESTION 4
5 - QUESTION 5
Press Option (1 - 5) to execute the program and press 0 to Exit
Enter Your Option:
2
-----> EXECUTING QUESTION - 2 <-----
Enter two Characters :
I2
Information Technology Sophomore
Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu
Enter your choice :
0
```

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OUTPUT

```
-----> MAIN MENU (ASSIGNMENT - 2) <-----  
1 - QUESTION 1  
2 - QUESTION 2  
3 - QUESTION 3  
4 - QUESTION 4  
5 - QUESTION 5  
Press Option (1 - 5) to execute the program and press 0 to Exit  
Enter Your Option:  
3  
-----> EXECUTING QUESTION - 3 <-----  
Plate Number : ODU1063  
Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu  
Enter your choice :  
0
```

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OUTPUT

```
-----> MAIN MENU (ASSIGNMENT - 2) <-----
1 - QUESTION 1
2 - QUESTION 2
3 - QUESTION 3
4 - QUESTION 4
5 - QUESTION 5
Press Option (1 - 5) to execute the program and press 0 to Exit
Enter Your Option:
4
-----> EXECUTING QUESTION - 4 <-----
Enter A Decimal Integer :
21
Binary Form = 10101
Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu
Enter your choice :
0
```

LAB ASSIGNMENT 2

OUTPUT

```
-----> MAIN MENU (ASSIGNMENT - 2) <-----
1 - QUESTION 1
2 - QUESTION 2
3 - QUESTION 3
4 - QUESTION 4
5 - QUESTION 5
Press Option (1 - 5) to execute the program and press 0 to Exit
Enter Your Option:
5
-----> EXECUTING QUESTION - 5 <-----
----- QUIZ -----
1. The Earth's average temperature has increased over the past century.
a. True      b. False      c. Not sure      d. None
a
2. The main cause of global warming is deforestation.
a. True      b. False      c. Not sure      d. None
b
3. Renewable energy sources can help reduce global warming.
a. True      b. False      c. Not sure      d. None
c
4. Global warming effects are only temporary.
a. True      b. False      c. Not sure      d. None
d
5. The polar ice caps are melting due to global warming.
a. True      b. False      c. Not sure      d. None
a

Your score: 3 out of 5
Time to brush up on your knowledge of global warming.
Some helpful websites to learn more:
- https://climate.nasa.gov/
- https://www.epa.gov/climate-change
- https://www.ipcc.ch/
- https://climate.nationalgeographic.com/
Do you want to execute this Question Again then Press (y) Otherwise Press 0 (b) to go to main menu
Enter your choice :
0
```

LAB ASSIGNMENT 2

OUTPUT

```
-----> MAIN MENU (ASSIGNMENT - 2) <-----  
1 - QUESTION 1  
2 - QUESTION 2  
3 - QUESTION 3  
4 - QUESTION 4  
5 - QUESTION 5  
Press Option (1 - 5) to execute the program and press 0 to Exit  
Enter Your Option:  
0  
Exiting The Program  
  
C:\Users\shehyar\Desktop\assignment 2 pf>_
```